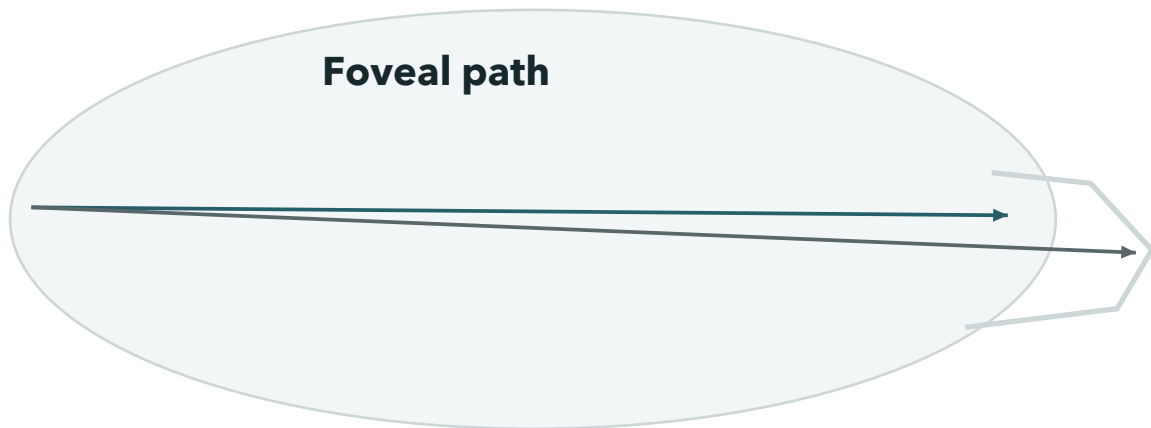


Complex biometry and IOL prediction

Part 2 | A formula cannot rescue the wrong measurement

Inspect input quality before comparing predicted implant powers.

Fixation and retinal endpoint must agree.



Deepest ectatic wall may be off axis

Original section: the foveal optical path can differ from the deepest staphyloma path.

In long eyes, confirm fixation, retinal endpoint and trace consistency. Optical and ultrasound measurements may disagree for geometrical reasons, not merely instrument calibration. Check corneal power, axial length and intended target together. An unexplained outlier should prompt reassessment rather than automatic averaging or selection of the longest number.

Previous refractive surgery changes assumptions

Corneal-power estimation and lens-position prediction can both be affected.

Changed feature	Potential consequence
Anterior/posterior curvature relationship	Usual keratometric index may misrepresent net power
Sampling area versus treated zone	Measured value may miss relevant central optics
K-dependent lens-position prediction	A second source of calculation error
Missing historical records	Use a validated no-history strategy

Use contemporary post-refractive methods appropriate to the actual procedure. Total-corneal measurements can add useful information but still require quality control. A correction developed for one operation is not automatically valid for another, and a perfect calculation does not remove irregular optical quality.

RK has geometry and time-dependent variability

A small central zone and changing refraction require repeated assessment.

RK feature	Planning implication
Very flat small central optical zone	Routine sampling can misrepresent central power
Diurnal change	One reading may be unrepresentative
Long-term refractive drift	Discuss prediction uncertainty
Prior incisions	Consider wound and astigmatic effects

Review repeated keratometry or tomography and refraction under comparable conditions. Use a validated post-RK calculation approach and counsel about residual error. RK should not simply be entered as post-myopic LASIK: the altered surface relationships and small-zone sampling problems differ.

Ectasia challenges both measurement and the target

Reproducible central optics matter more than the largest K value.

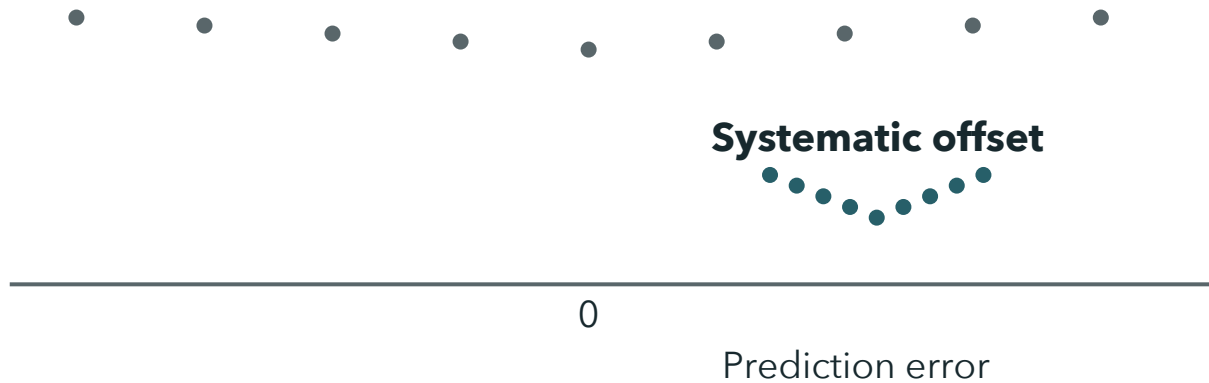
Assessment	Why it matters
Disease stability	Progression can undermine an implanted target
Regular central cylinder	May permit predictable toric correction
Irregularity and contact-lens dependence	Limits spectacle-only expectations
Repeatable lens-free measurements	Warping can distort planning

Choose a calculation strategy that recognizes abnormal corneal geometry. The desired refraction should also account for later contact-lens rehabilitation and the fellow eye. An apparently precise predicted power does not mean that all higher-order aberrations will be corrected by the IOL.

Prediction error is an audit signal

Separate systematic bias from random spread.

Broad spread



Original conceptual distribution: a mean offset and broad scatter imply different corrections.

Compare stable postoperative refraction with the intended target using a consistent sign convention. A systematic shift may suggest constant optimization or measurement bias; wide scatter may suggest variable inputs or lens position. Do not adjust every patient from one unusual result. Record formula, implant, measurement quality and operative events so the review can identify a cause.